

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It is not necessary to check that $A^{-1}A=I$ since A is invertible. $\square$

Another View of Matrix Inversion

Denote the columns of I_n by $e_1, \dots, e_n$. Then row reduction of $[A I]$ to $[I A^{-1}]$ can be viewed as the simultaneous solution of the n systems

$$Ax = e_1, Ax = e_2, \dots, Ax = e_n \quad (2)$$

where the "augmented columns" of these systems have all been placed next to A to form $[A \ e_1 \ e_2 \ \dots \ e_n] = [A \ I]$.

The equation $AA^{-1} = I$ and the definition of matrix multiplication show that the columns of A^{-1} are precisely the solutions of the systems in (2).

Practice Problems

1) Use determinants to determine which of the following matrices are invertible.

a) $\begin{bmatrix} 3 & -9 \\ 2 & 6 \end{bmatrix}$

Determinant: $3(6) - (-9)(2) = 36 \neq 0$

$\therefore$ By Theorem 4, the matrix is invertible.

b) $\begin{bmatrix} 4 & -9 \\ 0 & 5 \end{bmatrix}$

Determinant: $4(5) - (-9)(0) = 20 \neq 0$

By Theorem 4, the matrix is invertible.

c) $\begin{bmatrix} 6 & -9 \\ -4 & 6 \end{bmatrix}$

Determinant: $6(6) - (-9)(-4) = 0$

By Theorem 4, the matrix is non-invertible.

2) Find the inverse of the matrix $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$, if it exists.

Solution:

$$[A I] = \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array} \right]$$

Row 3 = Row 3 + Row 1

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array} \right]$$

Row 3 = Row 3 - 2x Row 2

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 0 & 0 & -7 & -2 & 1 \end{array} \right]$$

Row 2 = Row 2 + Row 1

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 0 & 0 & -7 & -2 & 1 \end{array} \right]$$